Anuj Marudachalam

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ABOUT

HPC Engineer with hands-on experience in Linux systems engineering, custom OS development (Linux From Scratch), and HPC infrastructure deployment on PARAM Shavak systems. Skilled in cluster management, Slurm/Kubernetes orchestration, and VR-GPU computing stacks. Passionate about high-performance computing, systems optimization, and emerging AI infrastructure.

TECHNICAL SKILLS

High Performance Computing & Linux: LDAP, SLURM, PXE Boot, xCAT, Cluster Management, RHEL-based Linux (Rocky, AlmaLinux), Debian-based Linux (Ubuntu)
Programming & Automation: Python, C, Bash Scripting, Ansible
Virtualization & Containers: VMware, Proxmox, Kubernetes, Docker
Monitoring & Infrastructure: Prometheus, Grafana, Server Troubleshooting, RAID, VR Setup

WORK EXPERIENCE

Project Engineer — Full-Time

Sep 2024 – Present

Centre for Development of Advanced Computing (C-DAC), Pune, India

- Custom Linux OS Development (Linux From Scratch): Built and Developed a complete Linux operating system from source using LFS with complete Linux toolchain, compiled the kernel, integrated graphical environments, and configured system services, storage, and networking; troubleshooting low-level boot and package issues.
- Worked with graphical environments and GUI troubleshooting on Linux systems.
- PARAM Shavak HPC System Integration & Testing: Configured and integrated custom Linux environments, storage, and networking; validated hardware/software compatibility and conducted hardware and software testing on PARAM Shavak servers.
- VR Shavak Software Stack Development: Configured and tested VR software environments on Linux-based HPC systems with OpenXR support, SteamVR, and Blender VR.
- PARAM Shavak Deployment: Provisioned and configured 80+ PARAM Shavak servers with custom OS images, applications, and tooling to support deployment across AI, HPC, and Quantum simulation variants.
- AI Factory Study: Researched AI Factory and GPU server architectures, focusing on configuration and deployment strategies for future infrastructure.
- Kubernetes + Slurm Scheduling: Implemented an optimized Kubernetes + Slurm scheduling architecture using the SLINKY Slurm Operator and Slurm Bridge on a test server bed, exploring hybrid container-orchestration/HPC scheduling models.
- Created the PARAM Vidya test bed, a ready-to-deploy AI lab combining CPU/GPU compute, with nodes provisioned over PXE boot, to test the platform's software configuration (currently in testing).
- Represented C-DAC with a live demonstration of PARAM Shavak at TechVerse 2025 (MeitY), highlighting system design and applications in indigenous HPC.
- Represented C-DAC at Supercomputing India 2025 (SCI 2025): delivered a poster presentation on “Bharat Operating System Solutions (BOSS) for PARAM Shavak Desktop HPC System,” and hosted a PARAM Shavak user meet to showcase the platform's newest features and capabilities.
- Showcased PARAM Shavak 2.0 and demonstrated HPC capabilities for AI applications at NSM (National Supercomputing Mission) Conclave 2026 and at AI-Summit 2025.

Associate Software Engineer Intern

Feb 2023 – May 2023

Sieva Networks Solutions, Trivandrum, India

- Worked as a PHP Software Developer; performed web-testing for the Electronic Logging Device (ELD) project and integrated PostgreSQL database management.

EDUCATION

PG-Diploma in High Performance Computing — 79.44%, Grade A
Centre for Development of Advanced Computing, Pune, Maharashtra | Aug 2024

Master's in Computer Applications — CGPA: 7.96
Amrita Vishwa Vidyapeetham, Amritapuri Campus, Kerala | Jul 2023

B.Sc. Computer Systems and Design — CGPA: 7.49
PSG College of Technology, Coimbatore, Tamil Nadu | Jul 2021